Question 6 Test 2

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Theoretical Framework

This simulation models the evolution of a student's level of pain over time using a **discrete-time Markov Chain (DTMC)**. The pain level can evolve among three categorical states:

- **State 1**: Severe pain
- **State 2**: Moderate pain
- **State 3**: Mild pain

Let $\{X_n\}_{n\geq 0}$ be a time-homogeneous Markov Chain defined on the finite state space $S = \{1, 2, 3\}$, with transition probabilities defined by the matrix:

$$P = \begin{bmatrix} 1-p & p & 0 \\ p & 0 & 1-p \\ 0 & p & 1-p \end{bmatrix} \quad \text{where} \quad p = \frac{2}{5}$$

This structure satisfies the **Markov property**: the probability of transitioning to the next state depends only on the current state.

Simulation Justification

Transitions are simulated using uniform random variables $r \sim U(0,1)$, selecting the next state j such that:

$$\sum_{k=1}^{j-1} P_{ik} < r \leq \sum_{k=1}^{j} P_{ik}$$

This approach is equivalent to drawing from a **categorical distribution**:

$$X_{n+1} \sim \text{Categorical}(P_{X_n,\cdot})$$

Because the Markov chain is finite and time-homogeneous, its simulation via random sampling from the row-distribution of P is justified.

Run Simulation (Markov Chain ref Dr Wanduku)

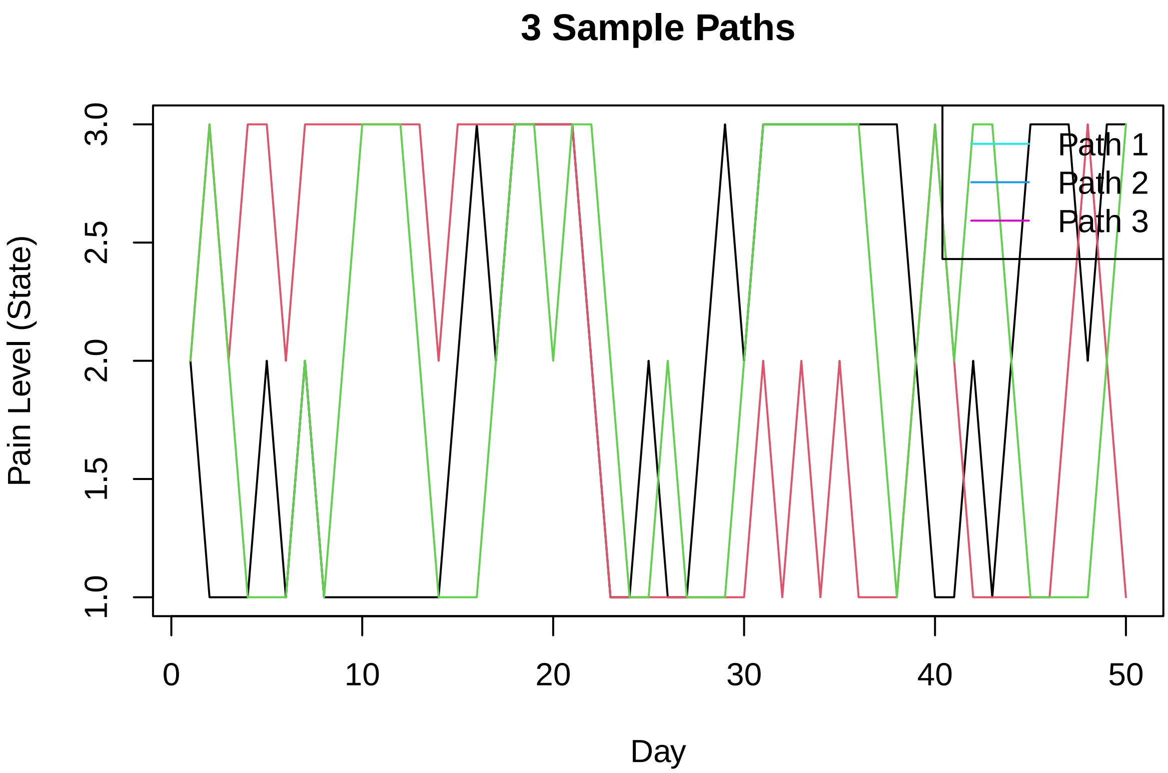
```r
set.seed(123)

p <- 2/5
time <- 60  # simulate for 60 days
nsim <- 500 # number of sample paths
initialcap <- 2 # start at state 2

X <- matrix(0, nrow = time, ncol = nsim)
Y <- matrix(0, nrow = time, ncol = nsim)
X[1, ] <- initialcap
for (i in 1:(time - 1)) {
  for (j in 1:nsim) {
    Y[i + 1, j] <- runif(1)
    if (X[i, j] == 1) {
      if (Y[i + 1, j] < 1 - p) {
        X[i + 1, j] <- 1
      } else {
        X[i + 1, j] <- 2
      }
    } else if (X[i, j] == 2) {
      if (Y[i + 1, j] < p) {
        X[i + 1, j] <- 1
      } else {
        X[i + 1, j] <- 3
      }
    } else if (X[i, j] == 3) {
      if (Y[i + 1, j] < p) {
        X[i + 1, j] <- 2
      } else {
        X[i + 1, j] <- 3
      }
    }
  }
}
```

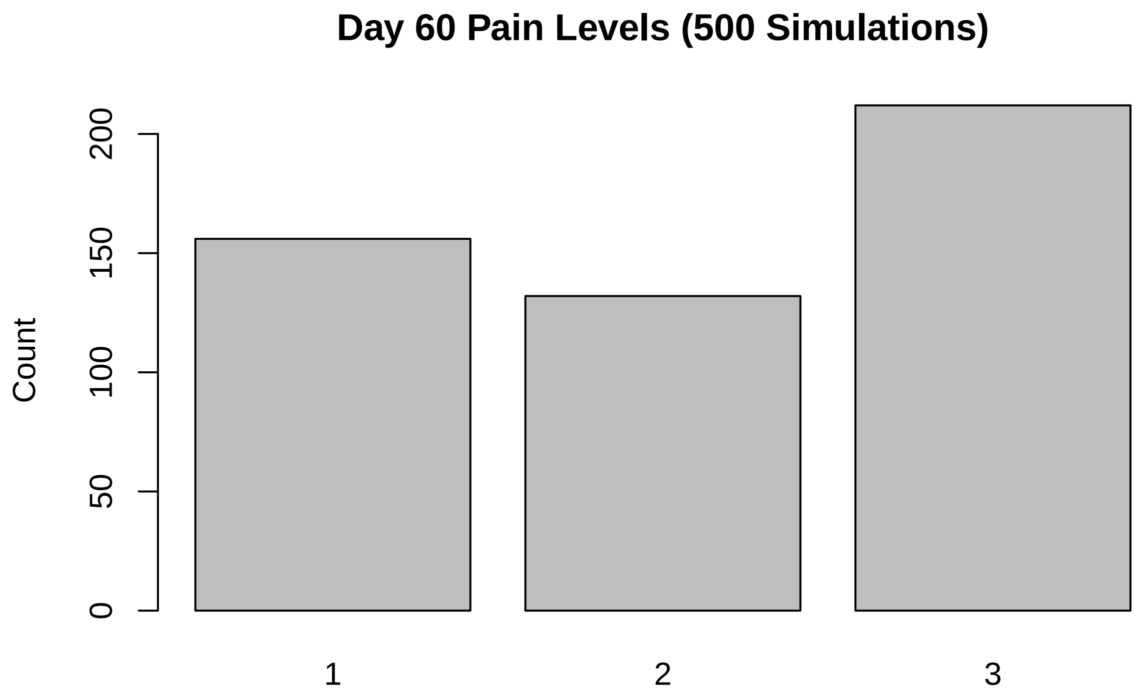
Part (a): Produce a graph of three sample paths of the process starting in state "2", and over 50-time steps. Select a given path, and describe the student's progression in pain, over the 50 days.

```r
matplot(1:50, X[1:50, 1:3], type = "l", col = 1:3, lty = 1,
        ylab = "Pain Level (State)", xlab = "Day", main = "3 Sample Paths")
legend("topright", legend = c("Path 1", "Path 2", "Path 3"), col = c(5,4,6), lty = 1)
```



Part (b): Use 500 sample paths to generate a sample from the population on the 60th day. Produce a graphical summary of the sample, describe the data, and answer the next question.

```r
final_states <- X[time, ]
barplot(table(final_states), col = "gray",
        main = "Day 60 Pain Levels (500 Simulations)", ylab = "Count")
```



Interpretation: the dataset consists of the pain level states of 500 students on the 60th day of recovery, each generated from a separate simulated Markov Chain path starting at State 2. The data are categorical, representing three possible outcomes: State 1 (severe pain), State 2 (moderate pain), and State 3 (mild pain). The bar chart shows that the most frequently observed state on Day 60 was State 3, indicating that a substantial proportion of students had recovered to mild pain. State 1 was the second most frequent, while State 2 had the lowest count. This distribution reflects the variability in recovery outcomes and provides a snapshot of likely pain levels among students after 60 days of progression through the Markov process.

Part (c): Observe that the "levels of pain" is a categorical variable, thus, to conduct inferences in the population on the 60th day, we must select an appreciate numeric measure of center such as the mode or proportion. Use a 95% confidence interval (CI) to describe how the student would feel on the 60th day. That is, find the CI, and interpret the result.

To describe how a student is likely to feel on the 60th day, we compute the **95% confidence intervals** for the proportions in each pain state. Since the data are categorical and represent proportions, we use the **normal approximation to the binomial distribution**:

$$\hat{p} \pm z_{\alpha/2} \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Where: - $\hat{p}$ is the sample proportion, - $z_{\alpha/2} = 1.96$ for a 95% CI, - $n = 500$ is the number of simulations.

```
prop_state <- prop.table(table(final_states))
ci_95 <- function(p, n) {
  z <- qnorm(0.975)
  err <- z * sqrt(p * (1 - p) / n)
```

```r
  c(p - err, p + err)
}

ci_matrix <- t(sapply(prop_state, function(p) ci_95(p, nsim)))
rownames(ci_matrix) <- paste("State", names(prop_state))
colnames(ci_matrix) <- c("Lower 95%", "Upper 95%")
round(ci_matrix, 4)
```

```
##         Lower 95% Upper 95%
## State 1    0.2714    0.3526
## State 2    0.2254    0.3026
## State 3    0.3807    0.4673
```

Interpretation: Based on the simulation of 500 independent recovery paths, each representing a student's daily progression of pain over 60 days starting from moderate pain, we estimate with 95% confidence that between 38.1% and 46.7% of students will be in mild pain (State 3) by Day 60, indicating a high likelihood of significant recovery. In contrast, between 27.1% and 35.3% are expected to still experience severe pain (State 1), while 22.5% to 30.3% will remain in moderate pain (State 2). These results suggest that although most students are likely to improve to a mild pain state, a considerable portion may still experience discomfort, highlighting the variability in recovery outcomes. The relatively narrow confidence intervals add credibility to these estimates.